

Investor Composition, Bond Liquidity, and Labor Market

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Preliminary

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Motivation

Two important aspects of the corporate bond market:

- Bond market liquidity has real effects on firms, including employment.
- Bond investor base has shifted dramatically: mutual funds' share has roughly tripled over three decades and unlike buy-and-hold insurers, they offer daily redemption and sell into stress.

⇒ **Who holds a firm's bonds** may determine how hard liquidity shocks hit its workers.

This paper asks:

- ① Does **investor composition** shape how liquidity risk premium affects the labor market?
- ② Can a heterogeneous-firm, heterogeneous-investor model rationalize and quantify it?

This Paper

Empirics: panel of U.S. firms matched to their bondholder base (eMAXX), 2004–2022

- Firms' bond liquidity premium negatively predicts employment growth, investment, and debt issuance.
- The effect is **amplified by mutual fund ownership**: average MF ownership raises the sensitivity of employment growth to liquidity risk by $\sim 68\%$.
- **Causal**: investment-mandate-induced maturity-cutoff IV.
Placebo: no effect for insurance ownership — the channel is specific to **run-prone** investors.

Model (preliminary): heterogeneous firms, defaultable debt, aggregate shocks to productivity *and* bond market liquidity

- Transmission: liquidity dries up $\Rightarrow$ bond proceeds fall $\Rightarrow$ external financing tightens $\Rightarrow$ firms cut hiring.
- Investor composition governs exposure: firms held by **liquidity-sensitive** (mutual-fund-type) investors face larger employment responses.

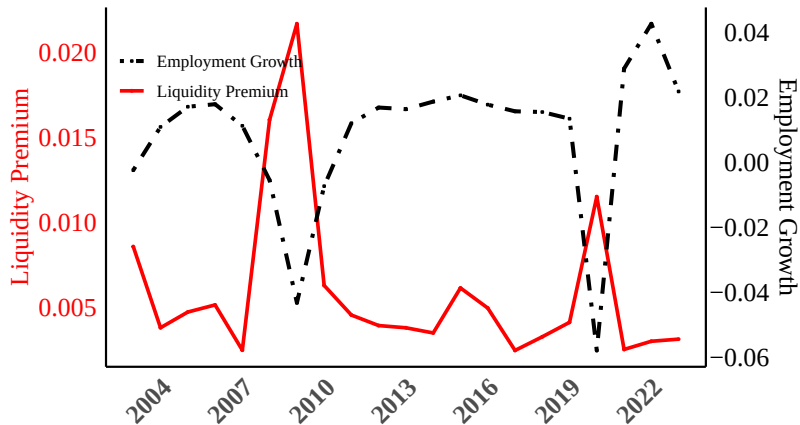
Roadmap

① Data and Empirical Results

② Model with Heterogeneous Firms

③ Conclusion

Liquidity Risk and Employment Growth Move Together



Correlation = -0.78 : liquidity dries up exactly when employment collapses — 2008–09, 2020.

vs. Unemployment Rate

Sample: U.S. nonfinancial firms with traded bonds and CDS, 2004–2022, annual

- **Key variable 1 — liquidity premium:** bond yield spread (TRACE) – CDS spread (Markit), à la Longstaff–Mithal–Neis (2005)
 - strips out default compensation $\Rightarrow$ isolates the liquidity component
- **Key variable 2 — bond ownership:** mutual fund and insurer holdings of each firm's bonds, from eMAXX
 - security-level holdings of buy-side institutions, aggregated to the firm
- **Firm outcomes:** employment, investment, debt issuance (Compustat)
- **Other:** macro and labor aggregates (FRED, BLS); fund flow shocks (Dou–Kogan–Wu 2025, WRDS)

From the Aggregate Link to the Firm Level

Empirical design: does the liquidity–employment link run through *who holds the bonds*?

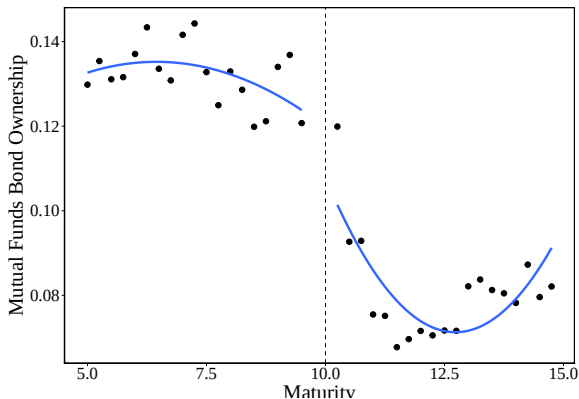
$$\Delta \text{Emp}_{i,t} = \beta_1 \underbrace{\text{Liq. Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1}}_{\text{transmission channel}} + \beta_2 \underbrace{\text{Liq. Premium}_{i,t-1}}_{\text{direct exposure}} + \beta_3 \text{Bond Ownership}_{i,t-1} + X'_{i,t-1} \beta + \gamma_i + \eta_t + \varepsilon_{i,t}$$

- $\beta_2 < 0$: firm-level analogue of the aggregate fact × Fin. Constraints, Recession
- $\beta_1 < 0$: **mutual fund ownership amplifies** transmission — our object of interest
- Identification challenge: ownership is endogenous $\Rightarrow$ mandate-cutoff IV

IV Construction: The 10-Year Maturity Cutoff

Construction of instrumental variable:

- Literature documents that a large fraction of corporate bond funds are intermediate-term, and only invest in bonds whose maturity is **less than** 10 years due to their investment mandates.
- Due to the investment mandate, bonds with time-to-maturity less than 10 years face **larger demand from mutual funds**, which are relatively short-term investors, hence the bond ownership becomes large.



IV Construction: The Mandate IV (MIV)

- Exploiting this investment-mandate-induced 10-year maturity cutoff, we follow Bretscher, Schmid, Ye (2025) to construct the instrumental variable MIV.

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$$MIV_{it} = \log \left(\sum_{h=1}^2 \frac{A_{ht}}{N_{ht}} \cdot \mathbb{1}_{ith} \right), \quad (1)$$

A_{ht} is the aggregate amount of assets benchmarked to maturity category h in month t ,

N_{ht} is the total number of bonds outstanding in maturity category h at time t .

- Aggregate MIV from the bond level to the firm level, using the amount outstanding of each bond as the weight:

$$MIV_{it}^{firm} = \frac{\sum_{j \in \mathcal{B}_{it}} AmtOut_{jt} \cdot MIV_{jt}}{\sum_{j \in \mathcal{B}_{it}} AmtOut_{jt}}. \quad (2)$$

Mutual Fund Ownership Amplifies the Liquidity Effect

$$\Delta \text{Employment}_{i,t} = \beta_1 \text{Liquidity Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1} + \beta_2 \text{Liquidity Premium}_{i,t-1} + \beta_3 \text{Bond Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Regression Fund Category	OLS	OLS Mutual	IV Mutual
Liquidity Premium	-0.5973** (-2.47)	-0.4630* (-1.88)	-0.4111* (-1.66)
Bond Ownership		0.0060 (0.47)	-0.0467 (-0.78)
Liquidity Premium $\times$ Bond Ownership		-1.1314** (-2.09)	-1.6493** (-2.13)
No. of Observations	5,964	5,964	5,964
First-stage F-stat			35.44
Firm Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Clustering	Firm	Firm	Firm

An average level of mutual fund bond ownership (0.169) increases the sensitivity of employment growth to liquidity risk by $\frac{-0.4111 + (-1.6493 \times 0.169)}{-0.4111} - 1 = 67.8\%$.

Placebo: Insurance Fund Ownership

Placebo tests using insurance fund bond ownership:

- Literature shows that insurance funds are more stable, longer-horizon bond holders (Li, 2025; Zhou, 2025).
- Insurance fund bond ownership does not exhibit discontinuity regarding maturity.
- We use insurance fund bond ownership in the place of mutual fund bond ownership, and conduct the placebo tests.

Placebo Results: No Effect for Insurance Ownership

$$\Delta \text{Employment}_{i,t} = \beta_1 \text{Liquidity Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1} + \beta_2 \text{Liquidity Premium}_{i,t-1} + \beta_3 \text{Bond Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Regression Fund Category	OLS	OLS Mutual	IV Mutual	OLS Insurance	IV Insurance
Liquidity Premium	-0.5973** (-2.47)	-0.4630* (-1.88)	-0.4111* (-1.66)	-0.5994** (-2.44)	-1.2570 (-1.60)
Bond Ownership		0.0060 (0.47)	-0.0467 (-0.78)	0.0041 (0.48)	-0.1335 (-0.83)
Liquidity Premium $\times$ Bond Ownership		-1.1314** (-2.09)	-1.6493** (-2.13)	0.1460 (0.34)	4.9070 (0.98)
No. of Observations	5,964	5,964	5,964	5,452	5,452
First-stage F-stat			35.44		0.63
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm

Same Pattern for Investment and Debt Issuance

$$Y_{i,t} = \beta_1 \text{Liquidity Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1} \\ + \beta_2 \text{Liquidity Premium}_{i,t-1} + \beta_3 \text{Bond Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Dependent Variable	Investment			Debt Issuance		
Fund Category Regression	Mutual OLS	Mutual IV	Insur. OLS	Mutual OLS	Mutual IV	Insur. OLS
Liquidity Premium	-0.0970 (-1.11)	-0.0214 (-0.19)	-0.2755*** (-2.71)	-0.3399*** (-2.74)	-0.3322** (-2.19)	-0.4718** (-2.55)
Bond Ownership	-0.0047 (-0.50)	-0.0147 (-0.64)	-0.0024 (-0.36)	-0.0393*** (-2.67)	-0.0821** (-2.02)	0.0205 (1.60)
Liquidity Premium × Bond Ownership	-0.7532* (-1.82)	-1.3187* (-1.78)	0.4994 (1.58)	-0.7302** (-2.15)	-0.8357* (-1.75)	1.2710* (1.90)
First-stage F-stat		103.61	0.63 [†]		139.48	0.72 [†]
No. of Observations	5,997	5,997	5,481	6,123	6,123	5,580
Firm & Time FE	Yes	Yes	Yes	Yes	Yes	Yes

The Liquidity Interaction Is Not Proxying for Other Channels

$$\Delta \text{Emp}_{i,t} = \beta_1 \text{Liq. Premium}_{i,t-1} \times \text{Ownership}_{i,t-1} + \beta_2 \text{Channel}_{i,t-1} \times \text{Ownership}_{i,t-1} \\ + \beta_3 \text{Liq. Premium}_{i,t-1} + \beta_4 \text{Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Dependent Variable	Δ Employment		
Competing channel	Fund Flow Shock	Rollover Risk	Cash Saving
Liquidity Premium	-0.3751* (-1.83)	-0.4506* (-1.77)	-0.4262* (-1.71)
Liquidity Premium × Bond Ownership	-1.7310*** (-2.86)	-1.4320* (-1.81)	-1.3570* (-1.78)
Channel × Bond Ownership	0.0047* (1.92)	-0.4210* (-1.81)	0.0774 (0.20)
First-stage F-stat	30.40	13.27	29.32
No. of Observations	5,816	5,922	5,922
Firm & Time FE	Yes	Yes	Yes

Takeaway: β_1 remains negative and significant when each competing interaction is added. Similar for investment and debt issuance.

All outcomes

Roadmap

- ① Data and Empirical Results
- ② **Model with Heterogeneous Firms**
- ③ Conclusion

Model Overview

Goal: link firm employment decisions to bond market liquidity and investor composition.

Environment: discrete time infinite horizon

- Heterogeneous firms with idiosyncratic productivity shocks and aggregate shocks.
- Aggregate shock X_t drives both productivity and bond market **liquidity**.
- Firms choose **investment** I_t , **labor** N_t , and **debt** B_{t+1} ; costly equity issuance.
- **One-period defaultable debt**; default when retraining value < 0 .
- Five state variables $S_t = (K_t, N_t, B_t, X_t, Z_t)$, can be solved with deep learning.

Today: reduced-form liquidity shock in bond pricing.

Next step: micro-found with a Duffie–Gârleanu–Pedersen OTC search model with heterogeneous investors.

Technology

Production function: decreasing returns to scale ($\theta < 1$)

$$Y_{it} = Z_{it} K_{it}^{\alpha} (X_t N_{it})^{(1-\alpha)\theta}$$

- Z_{it} : idiosyncratic productivity; X_t : aggregate productivity, labor-augmenting. Both log AR(1).
- Capital: $K_{i,t+1} = (1 - \delta)K_{it} + I_{it}$.
- Convex adjustment costs in capital *and* labor (scaled by the wage bill):

$$\Phi_{it} = \frac{c_k}{2} \left(\frac{I_{it}}{K_{it}} \right)^2 K_{it} + \frac{c_n}{2} \left(\frac{N_{i,t+1} - (1 - \delta_n)N_{it}}{N_{it}} \right)^2 W_t N_{it}$$

Debt and Equity Financing with Liquidity Wedge

Budget constraint (Firm's net payout before equity issuance cost):

$$E_t = (1 - \tau)(Y_t - W_t N_t - f) + \tau \delta K_t - I_t - \Phi_t - (1 + (1 - \tau)c)B_t + P_t B_{t+1} e^{\gamma x_t}$$

Interpretation:

- Liquidity wedge $e^{\gamma x_t}$ scales bond proceeds.
- In bad aggregate state (x_t low): $e^{\gamma x_t} \downarrow \Rightarrow$ effective $P_t \downarrow \Rightarrow$ financing cost $\uparrow$.
- γ measures sensitivity to liquidity:
 - high γ : held more by mutual fund $\Rightarrow$ strong liquidity effect
 - low γ : held more by insurer/pension $\Rightarrow$ stable funding

Dividends D_t and issuance cost Ψ_t :

$$D_t = E_t - \Psi_t, \quad \Psi_t = (\eta_0 - \eta_1 E_t) \mathbf{1}_{\{E_t < 0\}}$$

Wage (rigid, Belo–Lin 2014): $W_t = \tau_1 \exp(\tau_2 x_t)$, $0 < \tau_2 < 1$

procyclical, less volatile than output

Firm Problem

Firm chooses whether to default $\mathcal{D}_{t+1} \in \{0, 1\}$ **and** $\{K_{t+1}, N_{t+1}, B_{t+1}\}$:

$$V(S_t) = \max \left\{ 0, \max_{K_{t+1}, N_{t+1}, B_{t+1}} \left[D_t + \mathbb{E}_t M_{t,t+1} V(S_{t+1}) \right] \right\}$$

Taste-shock smoothing of default

Bond price:

$$P_t = \mathbb{E}_t \left[M_{t,t+1} \left((1 + c)(1 - \mathcal{D}_{t+1}) + \frac{L_{t+1}}{B_{t+1}} \mathcal{D}_{t+1} \right) \right]$$

Recovery:

$$L_t = \varphi \left[(1 - \tau)(Y_t - W_t N_t - f) + (1 - \delta)K_t \right]$$

Exogenous SDF (Berk et al, 1999; Zhang, 2005):

$$M_{t,t+1} = \beta \frac{e^{\gamma_t(x_t - x_{t+1})}}{\mathbb{E}_t[e^{\gamma_t(x_t - x_{t+1})}]}, \quad \gamma_t = \gamma_0 + \gamma_1 x_t$$

Calibration

Param.	Interpretation	Value	Param.	Interpretation	Value
β	Discount factor	0.9959	η_0	Equity issuance fixed cost	0.50
τ	Corporate tax rate	0.30	η_1	Equity issuance slope	0.0
θ	Returns to scale	0.82	τ_1	Wage scale	0.025
α	Capital share	0.28	τ_2	Wage elasticity to X_t	1.0
δ	Capital depreciation	0.01	ρ_x	Aggregate persistence	0.95
δ_n	Labor exit rate	0.0214	σ_x	Aggregate volatility	0.10
φ	Recovery rate	0.60	ρ_z, σ_z	Idio. persist./vol. (z)	0.97 / 0.50
c_k	Capital adjustment cost	10.0	$\bar{z}$	Avg. log idio. productivity	-2.6
c_n	Labor adjustment cost	7.5	ξ_d	Taste-shock scale (default)	0.10
f	Fixed operating cost	1.3	γ	Liquidity sensitivity (MF / insurer)	0.010 / 0.012

Monthly. Source: Belo et al 2025, Hennessy–Whited 2007, or matched to data moments. Internally disciplined: c_k, c_n, f, γ .

Model vs. data moments

Use Deep Learning to Solve the Model

5-dim. state (K, N, B, X, Z) with endogenous default $\mathcal{D}$ and bond price P . Nested fixed-point iteration over (V, P) .

- 1 Approximate V^r and the three policies by neural networks:

$$V^r(S) \approx \widehat{V}^r(S; \Theta_V), \quad K'(S) \approx \widehat{K}(S; \Theta_K), \quad N'(S) \approx \widehat{N}(S; \Theta_N), \quad B'(S) \approx \widehat{B}(S; \Theta_B).$$

NN architecture

- 2 Start from $\Theta^0 = (\Theta_V, \Theta_K, \Theta_N, \Theta_B)$. At iteration n :

- 1 Sample K state points $Q^n = \{S_k\}_{k=1}^K$ from the state space.
- 2 Form residuals of the **Bellman equation** and the three **FOCs** for $K_{t+1}, N_{t+1}, B_{t+1}$, and compute the average squared loss:

$$L(\Theta^n, Q^n) = \frac{1}{K} \sum_{k=1}^K \left[\underbrace{(V_k^r - D_k - \mathbb{E}MV_{k+1})^2}_{\text{Bellman}} + \sum_{x \in \{K, N, B\}} \underbrace{(LHS_k^x - RHS_k^x)^2}_{\text{FOC residual}} \right]$$

- 3 Update parameters by SGD: $\Theta^{n+1} = \Theta^n - \zeta^n \nabla_{\Theta} L(\Theta^n, Q^n)$.
- 4 Repeat until $L \leq \varepsilon$. Final residuals $\sim 10^{-7}$ – 10^{-6} across state space.
- 3 With $\widehat{V}^r$ and policies in hand, $\mathcal{D}$, P , and IRFs are implied by the equilibrium equations.

Results: Value Function

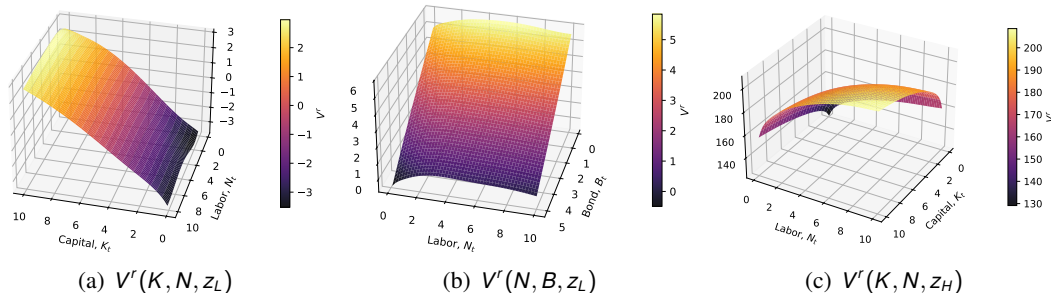
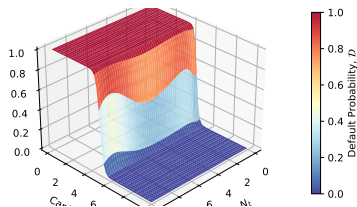


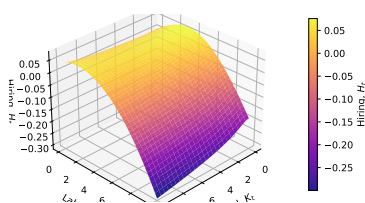
Figure: V^r at middle B or K , low ($z_L = -4.279$) or high ($z_H = 0.759$) productivity

- V^r increases in K , decreases in B ; non-monotone in N (diminishing returns + wage rigidity + adjustment costs).
- At high z , $V^r > 0$ everywhere $\Rightarrow$ no default; at low z , $V^r < 0$ for low- K /high- B firms $\Rightarrow$ endogenous default region.

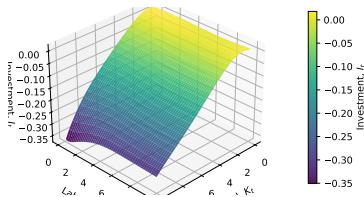
Results: Policy Functions of Low Productivity Firm



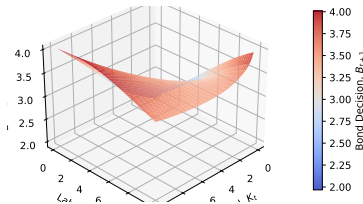
(a) Default D



(b) Hiring H



(c) Investment I



(d) Debt B'

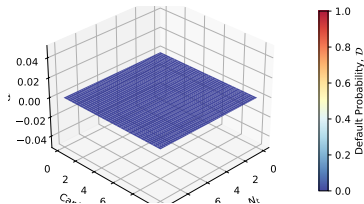
Policy as functions of (K, N) at middle B, x , low z .

Low- K firms optimally default.

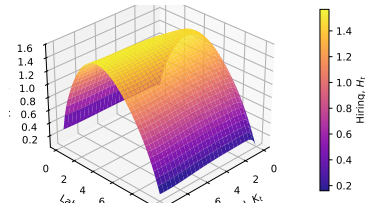
Survivors cut hiring and investment.

Higher current capital and employment $\Rightarrow$ lower hiring and investment (decreasing marginal returns).

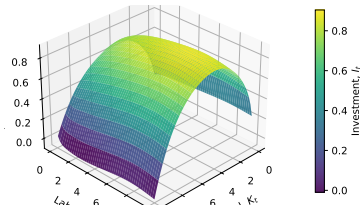
Results: Policy Functions of High Productivity Firm



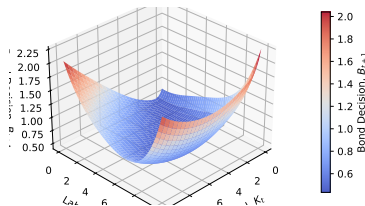
(e) Default D



(f) Hiring H



(g) Investment I



(h) Debt B'

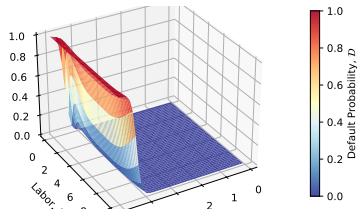
Policy as functions of (K, N) at middle B , x , high z .

Default disappears.

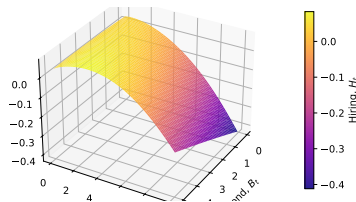
Hiring hump-shaped in N ;
investment hump-shaped in K
(convex adjustment costs).

Debt issuance U-shaped in
both K and N .

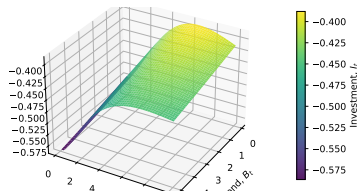
Results: Policy Functions in (N, B) Slice



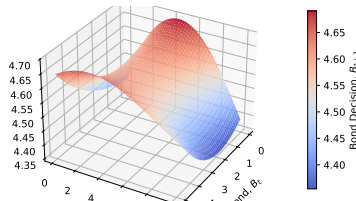
(i) Default D



(j) Hiring H



(k) Investment I



(l) Debt B'

Policy as functions of (N, B) at middle K , low x, z .

Default rises sharply with B .

Hiring decreases in N , increases in B (financing-needs channel).

Debt issuance hump-shaped in N , peaking at intermediate labor.

IRF of Employment to Liquidity Shock: Heterogeneous Investor

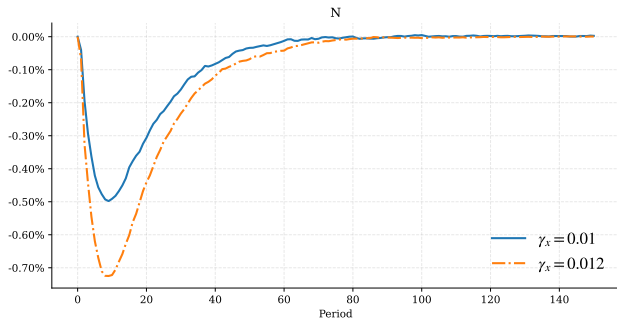


Figure: GIRF of aggregate employment N_t to a one-period adverse liquidity shock, by liquidity-sensitivity γ

- More elastic investor base $\Rightarrow$ larger employment $\downarrow$.

Roadmap

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Conclusion

What we find:

- Bond market liquidity shocks reduce firm employment growth — also investment and debt issuance.
- **Mutual fund ownership amplifies** transmission ($\sim 68\%$ at average ownership); **insurers show no effect** — identified with a mandate-induced maturity-cutoff IV.
- A heterogeneous-firm model with a liquidity wedge rationalizes the channel: liquidity dries up $\Rightarrow$ bond proceeds fall $\Rightarrow$ hiring falls.

Next step: micro-found the liquidity wedge with an OTC search model (Duffie–Gârleanu–Pedersen) with heterogeneous investors

- Endogenous, firm-specific liquidity premia from search and bargaining; investor composition as a firm-level state.
- Disciplines the liquidity-sensitivity parameter with micro structure rather than assuming it.

Literature

- Interaction of corporate bonds liquidity and default (He and Milbradt (2014), Chen et al. (2018), Chang (2023))
- Impact of financial frictions (excess bond premium) on real activities (Gilchrist and Zakrajšek (2012))
- Impact of labor market frictions on credit markets (Favilukis, Lin, and Zhao (2020))
- Role of institutional investor in shock transmission: Li and Yu (2021), Coppola (2025 RFS), Fang (2023), Zhou (2024), Li (2024)

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Aggregate Results: Liquidity Risk vs Unemployment Rate



Corporate bond liquidity risk vs unemployment rate: correlation = 0.49. [back](#)

Aggregate-Level Liquidity Risk vs Labor Market Dynamics

$$Y_t = \beta_1 \text{Liquidity Premium}_t + \mathbf{X}_t' \boldsymbol{\beta} + \varepsilon_t$$

Dependent Variable	Labor Market Tightness	Unemployed Persons Per Job Opening Ratio	Unemployment Rate	Employment Growth
Panel A				
Bid-Ask Spread	-0.8935* (-1.83)	2.9728*** (2.91)	2.9661** (2.26)	-0.0037* (-1.68)
Panel B				
Aggregate Liquidity Premium	-38.2147** (-2.04)	102.1171** (2.39)	95.7323* (1.73)	-0.2819*** (-2.73)
HAC Robust SE	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes

$\mathbf{X}_t'$: industry production growth, ΔVIX .

Liquidity Risk, Financial Constraints, and Economic Condition

Dependent Variable	Δ Employment		
	(1)	(2)	(3)
Liquidity Premium	-0.5973** (-2.47)	-2.1384** (-2.38)	-1.7601* (-1.93)
Financial Constraint		-0.0109 (-0.45)	-0.0083 (-0.34)
Liquidity Premium $\times$ Financial Constraints		-0.3674* (-1.84)	-0.3669* (-1.80)
Liquidity Premium $\times$ $\mathbb{1}(\text{Recession})$			-3.1134* (-1.71)
Liquidity Premium $\times$ Financial Constraints $\times$ $\mathbb{1}(\text{Recession})$			-0.7048* (-1.74)
No. of Observations	5,964	5,964	5,964
Firm Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Clustering	Firm	Firm	Firm

back

Robustness: 10-Year Tenor CDS Contracts

Dependent Variable	Δ Employment					Debt Issuance					Investment				
Regression	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV
Liquidity Premium	-0.4583*	-0.3038	-0.3149	-0.4601*	-0.9364	-0.3401***	-0.2455**	-0.2485**	-0.2255**	-0.8296*	-0.1802**	-0.0411	0.0537	-0.3214***	-0.5096
	(-1.89)	(-1.21)	(-1.24)	(-1.86)	(-1.32)	(-4.09)	(-2.34)	(-2.08)	(-2.24)	(-1.72)	(-2.07)	(-0.47)	(0.42)	(-2.84)	(-1.15)
Bond Ownership		0.0070	-0.0621	0.0026	-0.1815		-0.0346**	-0.0772*	0.0254**	-0.2100		-0.0079	-0.0231	-0.0016	-0.1401
		(0.56)	(-1.01)	(0.32)	(-1.13)		(-2.33)	(-1.86)	(2.07)	(-0.87)		(-0.90)	(-1.11)	(-0.24)	(-1.23)
Liquidity Premium × Bond Ownership		-1.256**	-1.371*	0.2497	3.708		-0.6238*	-0.6911*	0.4663	5.693		-0.9762**	-1.685**	0.6730**	1.588
		(-2.30)	(-1.85)	(0.61)	(0.77)		(-1.85)	(-1.67)	(0.93)	(1.37)		(-2.19)	(-2.15)	(2.06)	(0.65)
Channel		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance
No. of Observations	5,796	5,796	5,796	5,317	5,317	5,941	5,941	5,941	5,434	5,434	5,827	5,827	5,827	5,344	5,344
First-stage F -stat			39.70		0.52			172.31		1.62			82.83		1.31
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm

Robustness: Excluding Crisis Periods

Dependent Variable	Δ Employment					Debt Issuance					Investment				
Regression	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV
Liquidity Premium (LP)	-0.3949 (-1.42)	-0.2375 (-0.89)	-0.2273 (-0.86)	-0.4403 (-1.62)	-1.469 (-0.76)	-0.5280*** (-3.21)	-0.3076** (-2.10)	-0.2201 (-0.95)	-0.5704* (-1.96)	-2.806 (-0.65)	-0.1518* (-1.74)	-0.0895 (-0.87)	0.1315 (0.87)	-0.2346** (-2.08)	-0.6966 (-1.43)
Bond Ownership		0.0094 (0.72)	-0.0149 (-0.27)	0.0024 (0.27)	-0.0765 (-0.33)		-0.0287** (-2.01)	-0.0754* (-1.79)	0.0234* (1.65)	0.4376 (0.32)		-0.0053 (-0.58)	-0.0082 (-0.40)	-0.0001 (-0.07)	-0.1174 (-1.09)
Liquidity Premium × Bond Ownership		-1.4940** (-2.36)	-1.6480* (-1.87)	0.2756 (0.54)	8.8430 (0.64)		-1.1880** (-2.36)	-1.6900** (-2.00)	1.5660** (1.99)	25.4850 (0.55)		-0.4505 (-0.92)	-1.9750** (-2.23)	0.2142 (0.66)	2.5850 (0.99)
Channel		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance
No. of Observations	5,064	5,064	5,064	4,574	4,574	5,184	5,184	5,184	4,663	4,663	5,092	5,092	5,092	4,598	4,598
First-stage F -stat			34.73		0.22			131.93		0.14			71.23		1.28
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm

Appendix: Model vs. Data Moments

Moment	Data	Model
<i>Panel A: Firm-level moments</i>		
ΔDebt	0.038	0.004
$\text{SD}(\Delta\text{Debt})$	0.213	0.120
$\text{AR1}(\Delta\text{Debt})$	-0.126	0.867
$\Delta\text{Employment}$	0.010	0.005
$\text{SD}(\Delta\text{Employment})$	0.134	0.063
$\text{AR1}(\Delta\text{Employment})$	-0.011	0.974
Investment Rate	0.194	0.170
$\text{SD}(\text{Investment Rate})$	0.115	0.396
$\text{AR1}(\text{Investment Rate})$	0.307	0.982
<i>Panel B: Aggregate moments</i>		
Leverage Ratio	0.328	0.485
Default Probability	0.017	0.013
Credit Spread	0.011	0.017
Stock Return	0.089	0.120
Stock Return Volatility	0.156	0.142
Sharpe Ratio	0.581	0.483

Takeaway: default rate and leverage close; credit spread realistic; firm-level dynamics same order of magnitude. Persistence too high (smoothing from NN approximation); investment-rate vol too high — next-round calibration target.

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Approximate V^r and Policies by Neural Network (Feed-Forward, Fully Connected)

- Let $\omega = (K, N, B, X, Z)$. We approximate each of $\widehat{V}^r, \widehat{K}, \widehat{N}, \widehat{B}$ by an NN of the form

$$\mathbf{h}^{(1)} = \phi^{(1)}(\mathbf{W}^{(1)}\omega + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(\mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$$\vdots$$

$$\mathbf{h}^{(H)} = \phi^{(H)}(\mathbf{W}^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer } H$$

$$\text{output} = \sigma(\mathbf{h}^{(H)}) \quad \dots \widehat{V}^r, \widehat{K}, \widehat{N}, \text{ or } \widehat{B}$$

- Terminology:
 - H : number of *hidden layers*
 - Length of $\mathbf{h}^{(i)}$: number of *neurons* per hidden layer
 - $\phi^{(i)}$: *activation* for hidden layer i
 - σ : *activation* for the output layer
 - $\Theta = (\mathbf{W}^{(1)}, \dots, \mathbf{W}^{(H)}, \mathbf{b}^{(1)}, \dots, \mathbf{b}^{(H)})$: *parameters*, trained by SGD.

Appendix: Taste-Shock Smoothing of the Default Decision

Why: discrete default $\Rightarrow$ non-differentiable value $\Rightarrow$ breaks gradient-based deep-learning solution. Standard fix from sovereign-default literature (Dvorkin–Saprizo–Sosa-Padilla 2021; Chatterjee et al. 2023).

Setup. Add i.i.d. type-I EV shocks ϵ^r, ϵ^d with scale $\xi_d > 0$ to continuation value V^r and default value V^d . The ex-ante firm value and default probability are:

$$V = \mathbb{E}_\epsilon \max\{V^r + \epsilon^r, V^d + \epsilon^d\} = \xi_d \log \left[\exp \frac{V^r}{\xi_d} + \exp \frac{V^d}{\xi_d} \right], \quad \mathcal{D} = \frac{\exp(V^d/\xi_d)}{\exp(V^r/\xi_d) + \exp(V^d/\xi_d)}$$

Normalize $V^d = 0$:

$$V = \xi_d \log \left[1 + \exp \frac{V^r}{\xi_d} \right], \quad \mathcal{D} = \frac{1}{1 + \exp(V^r/\xi_d)}$$

Deterministic limit. As $\xi_d \rightarrow 0$: $\mathcal{D} \rightarrow \mathbf{1}_{\{V^r < 0\}}$, recovering the standard limited-liability default rule.

Calibration: $\xi_d = 0.10$.

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